

Week 5: Potential Energy and Energy Conservation – Tutorial Sheet

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Use the Master Procedure: identify, sketch the states and choose the reference level, sort the forces into U and W_{other} , predict, select the method, execute $K_1 + U_1 + W_{\text{other}} = K_2 + U_2$, evaluate. Use $g = 9.80 \text{ m/s}^2$. Problems A1 and A2 are solved in full; final answers to the others are at the end.

A. Worked examples**A1. A skier on a smooth slope**

A 60.0 kg skier starts from rest at the top of a smooth ($\mu = 0$) slope, 40.0 m above the valley floor. Find her speed at the bottom, and at the point 15.0 m above the valley floor. Does her mass matter?

Solution. Take $y = 0$ at the valley floor. Only gravity does work (the normal force is perpendicular to the motion), so $K_1 + U_1 = K_2 + U_2$ with $K_1 = 0$: $mg y_1 = \frac{1}{2}mv^2 + mg y_2$, hence $v = \sqrt{2g(y_1 - y_2)}$. Bottom: $v = \sqrt{2(9.80)(40.0)} = 28.0 \text{ m/s}$. At 15.0 m: $v = \sqrt{2(9.80)(25.0)} = 22.1 \text{ m/s}$. The mass cancels, so it does not matter.

A2. A block, friction and a spring

A 2.00 kg block slides at 3.00 m/s along a rough floor ($\mu_k = 0.20$). After 1.50 m it meets a light horizontal spring ($k = 400 \text{ N/m}$, the floor under the spring is also rough). Find the maximum compression of the spring.

Solution. Friction force $f_k = \mu_k mg = 0.20(2.00)(9.80) = 3.92 \text{ N}$. Take state 1 at the start ($K_1 = \frac{1}{2}(2.00)(3.00)^2 = 9.00 \text{ J}$) and state 2 at maximum compression x ($K_2 = 0$, $U_{\text{el}} = \frac{1}{2}kx^2$). The total sliding distance is $1.50 + x$: $K_1 + W_f = U_{\text{el},2} \Rightarrow 9.00 - 3.92(1.50 + x) = 200x^2$, i.e. $200x^2 + 3.92x - 3.12 = 0$. $x = \frac{-3.92 + \sqrt{3.92^2 + 4(200)(3.12)}}{2(200)} = 0.115 \text{ m}$. Check: spring energy $\frac{1}{2}(400)(0.115)^2 = 2.67 \text{ J}$; friction removed $3.92(1.615) = 6.33 \text{ J}$; total 9.00 J. ✓

B. Practice**B1. A stone in a vertical circle**

A 0.300 kg stone on a string of length 0.800 m moves in a vertical circle. Its speed at the lowest point is 7.00 m/s. Find (a) its speed and (b) the tension at the top of the circle. (c) What is the smallest speed at the bottom for which the string stays taut all the way round?

B2. A diver

A 65.0 kg diver steps off a 10.0 m platform. (a) Find her speed on reaching the water. (b) She stops 3.00 m below the surface. Find the average force the water exerts on her.

B3. A hanging spring

A 0.500 kg mass hangs at rest from a light vertical spring, which stretches 0.100 m. The mass is pulled down a further 0.150 m and released from rest. Find the spring constant and the speed of the mass as it passes the equilibrium position.

B4. Down a rough incline onto a spring

A 3.00 kg block is released from rest on a 35° incline ($\mu_k = 0.30$) and slides 2.00 m before it meets a light spring ($k = 800 \text{ N/m}$) lying along the incline. Find the maximum compression of the spring.

B5. Force and energy diagram

A 0.400 kg particle has potential energy $U(x) = \frac{4.0}{x^2} - \frac{8.0}{x}$ (joules, $x > 0$ in metres). (a) Find the force at $x = 0.50 \text{ m}$ and at $x = 2.0 \text{ m}$. (b) Find the equilibrium position and state whether it is stable. (c) For $E = -2.0 \text{ J}$, find the turning points and the greatest speed.

B6. Challenge: the method is not given

A small block is released from rest at a height of 2.00 m on a smooth track. At the bottom the track becomes a horizontal rough strip of length d ($\mu_k = 0.300$), followed by a smooth vertical loop of radius 0.500 m. What is the greatest d for which the block still goes all the way round the loop?

C. Concept questions

1. Why can the gravitational potential energy of a ball be negative? Is the work done by gravity changed by the choice of $y = 0$?
2. Two frictionless ramps of different slope reach the same height. A block released from rest at the top of each reaches the bottom with the same speed. Why is this not true if friction acts?
3. A person walks up a staircase at constant speed. Which energy transformations occur? Which force does the positive work on the person?
4. On an energy diagram the total energy line lies below a maximum of U . What does that say about the particle's motion?

Answers

- B1 (a) 4.20 m/s; (b) 3.7 N; (c) $v_{\min} = \sqrt{5gL} = 6.26 \text{ m/s}$
- B2 (a) 14.0 m/s; (b) $2.76 \times 10^3 \text{ N}$ (about 4.3 times her weight)
- B3 $k = 49.0 \text{ N/m}$; $v = 1.48 \text{ m/s}$ (equals $\sqrt{k/m} \times 0.150$)
- B4 $x = 0.232 \text{ m}$ (quadratic: $400x^2 - 9.64x - 19.3 = 0$)
- B5 (a) $F = +32 \text{ N}$ at 0.50 m (toward $+x$), $F = -1.0 \text{ N}$ at 2.0 m; (b) $x = 1.0 \text{ m}$, $U = -4.0 \text{ J}$, stable; (c) $x = 2 \pm \sqrt{2} = 0.59 \text{ m}$ and 3.41 m ; $v_{\max} = 3.16 \text{ m/s}$ at $x = 1.0 \text{ m}$
- B6 $d = 2.5 \text{ m}$ (needs $v_{\text{top}}^2 \geq gR$; speed at the bottom of the loop 4.95 m/s)
- C1 reference level is a choice; $W = -\Delta U$ is unchanged
- C2 friction depends on the path length; the steeper ramp is shorter
- C3 chemical $\rightarrow U_{\text{grav}}$ and thermal; the force of the stairs does no work on the person; the muscles supply the energy
- C4 $K = E - U < 0$ at the maximum, so the particle cannot pass: it is confined to one side